

Log Analysis Foundations

Technical Overview: Log Types, Format Structure, Baseline Profiling & Anomaly Detection

■ EXECUTIVE SUMMARY

Log analysis is the cornerstone of Security Operations (SOC), threat hunting, and system troubleshooting. By establishing normalized log formats and behavioral baselines, analysts can rapidly differentiate routine system actions from malicious activity or system failures.

1. Core Log Types & Sources

In a modern enterprise IT architecture, security analysts rely on telemetry aggregated from multiple operational layers. Below are the five fundamental log types, their primary sources, and typical event types:

Log Type	Primary Source / Origin	Key Events & Telemetry Captured
System Logs	Operating Systems (Windows Event Log, Linux /var/log/messages, syslog, macOS unified log)	System boot/shutdown, service status, driver events, hardware diagnostics, OS updates.
Security Logs	Authentication servers (Active Directory, PAM), Identity Providers (Okta), Endpoint Detection (EDR)	User authentications (successful/failed), privilege escalations, account lockouts, policy changes.
Application Logs	Software applications, databases, custom enterprise software, microservices	Unhandled exceptions, software errors, API calls, user database queries, transactional records.
Network Logs	Firewalls, Network Routers, Switches, VPN Gateways, IDS/IPS (Snort, Suricata)	Packet flow, allowed/blocked connections, IP route changes, VPN sessions, intrusion alerts.
Web Server Logs	HTTP/HTTPS Servers (Apache HTTPD, Nginx)	Web page requests, client IP addresses, HTTP status codes, referrer information.

2. Apache Web-Server Log Line Breakdown

Web server logs follow standardized formats such as NCSA Common Log Format (CLF) or Combined Log Format. Understanding the explicit syntax of each field allows analysts to parse, filter, and extract threat indicators.

Raw Log Entry Example:

192.168.1.105 - - [30/Aug/2026:14:32:10 +0500] "GET /index.html HTTP/1.1" 200 2326

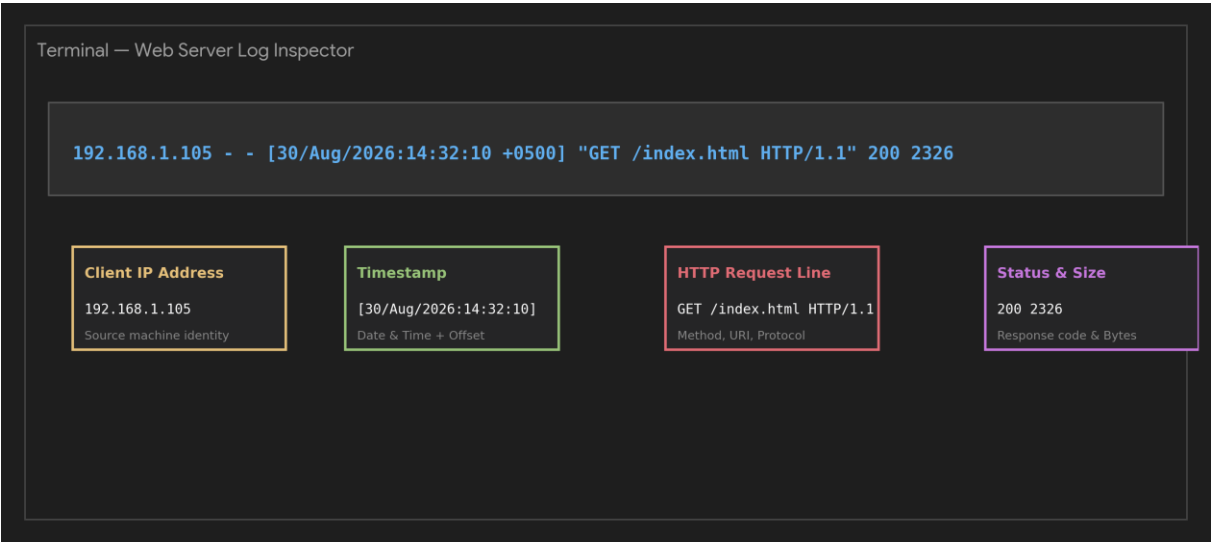


Figure 1: Visual Breakdown and Field Mapping of an Apache Combined Log Entry

Field Token	Extracted Value	Analyst Explanation & Purpose
Source IP Address	192.168.1.105	Identifies the network client making the request. Used for geolocation, reputation lookup, and rate-limiting.
Timestamp & Timezone	[30/Aug/2026:14:32:10 +0500]	Indicates exact date/time (UTC+5 offset). Crucial for timeline reconstruction and cross-log correlation.
HTTP Request Line	"GET /index.html HTTP/1.1"	Shows HTTP Method (GET), requested URI resource (/index.html), and

		protocol version. Helps detect command injection or path traversal.
HTTP Status & Bytes	200 2326	Status 200 (OK) indicates success. '2326' is payload size sent to client in bytes. Abnormally large sizes may indicate data exfiltration.

3. Comparative Analysis: Normal vs. Suspicious Log Entries

Detecting malicious behavior requires recognizing syntax deviations, illegal parameter strings, or unexpected HTTP status sequences.

NORMAL LOG ENTRY (Standard User Navigation)	SUSPICIOUS LOG ENTRY (SQL Injection / Path Traversal)
10.0.0.45 - - [30/Aug/2026:10:15:22 +0500] "GET /products/item.php?id=102 HTTP/1.1" 200 4520 "Mozilla/5.0 (Windows NT 10.0; Win64; x64)"	185.220.101.5 - - [30/Aug/2026:03:14:02 +0500] "GET /products/item.php?id=102%27%20UNION%20SELECT%20username,password%20FROM%20users-- HTTP/1.1" 500 240 "python-requests/2.28.1"
• Standard internal IP address (10.0.0.45) • Request occurred during normal business hours • Standard parameter lookup (id=102) • Standard browser User-Agent string	• Foreign/External public IP address • Unusual timestamp (03:14 AM off-hours) • Malicious Payload: URL-encoded SQL syntax (' UNION SELECT...) • Automated User-Agent: python-requests (scripted scanner)

4. Understanding 'Baselines' & Anomaly Detection

A baseline represents the mathematically modeled, normal operational behavior of an enterprise network, user base, or application over a given timeframe. Without an established baseline, security analysts cannot accurately identify anomalies, as there is no standard for comparison.

Key Components of a Baseline:

1. **Traffic Volume Metrics:** Average requests per minute/hour during business vs. non-business hours.
2. **User Authentication Patterns:** Typical login locations, active hours, and expected system roles.
3. **System Performance Profiles:** Standard CPU, memory, and database query latency bounds.

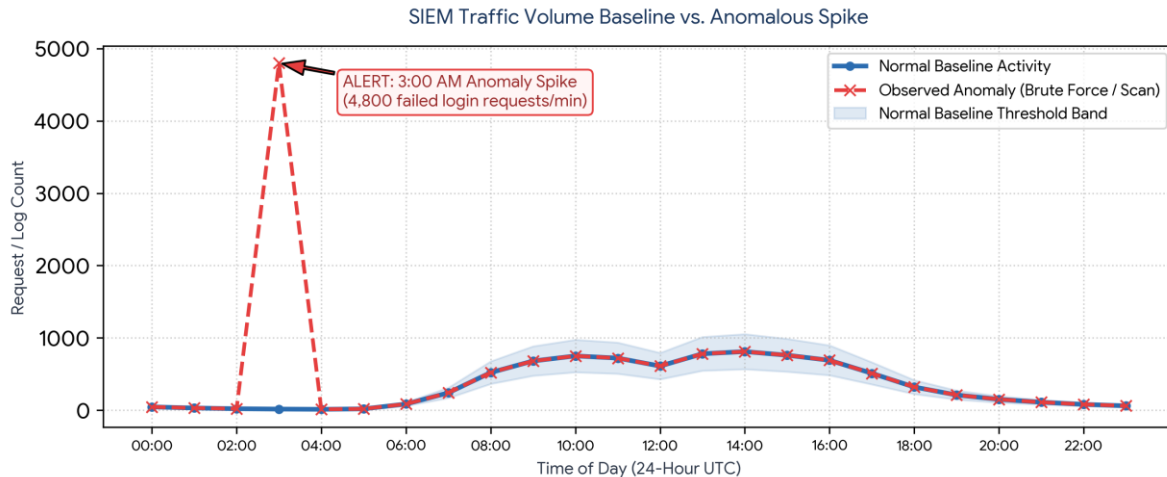


Figure 2: SIEM Telemetry Telemetry Metric — Detecting Off-Hour Traffic Anomaly against Established Baseline Band

How Analysts Spot Anomalies using Logs

- **Statistical Deviation:** Spikes in log volume exceeding +3 standard deviations from the moving average.
- **High-Frequency Failure Ratios:** A sudden influx of HTTP 404 (Not Found) or 401 (Unauthorized) errors indicating active vulnerability scanning or brute-force attempts.
- **Time-of-Day Anomalies:** Administrative actions (e.g., PowerShell execution, Active Directory changes) performed at 3:00 AM.
- **Signature & Pattern Matching:** Automated SIEM rules flagging regex signatures (e.g., `../`, `SELECT`, `<script>`) within input streams.

■ SUMMARY & BEST PRACTICES

Effective log analysis requires combining real-time automated SIEM correlation with analyst domain knowledge. Regularly updating baseline metrics ensures detection systems adapt to operational changes while maintaining high sensitivity to real threats.

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Intro to Logs

Learn the fundamentals of logging, data sources, collection methods and principles to step into the log analysis world.

   30 min  129,558 

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Task 1  Introduction

  Expanding Perspectives: Logs as Evidence of Historical Activity

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